<u>Number Programs</u> :

Question 1 :

Write a program to check if a number is a Special number or not. A special number is a number which is equal to the sum of the factorials of its digits.

For example, 145 is a special number. Sum of the factorials of the digits is equal to 145

Factorial of 1 $\rightarrow 1$
Factorial of 4 $\rightarrow 4 \times 3 \times 2 \times 1 = 24$
Factorial of 5 $\rightarrow 5 \times 4 \times 3 \times 2 \times 1 = 120$

1 + 24 + 120 = 145

Therefore, 145 is a special number.

Question 2 :

Write a program to check if a number is a Pronic number or not. A number is said to be a Pronic number if it is equal to the product of two consecutive integers.

For example, 6 is a Pronic number since it is equal to the product of two consecutive integers, 2 and 3.

$3 \times 4 = 12$
$5 \times 6 = 30$
$7 \times 8 = 56$

Here, 12, 30, and 56 are all Pronic numbers.

Question 3 :

Write a program to check if a number is a Harshad Number. A harshad number is a number which is divisible by the sum of its digits.

For example, 27 is a harshad number.

The sum of digits of 27 → 2 + 7 = 9
Since the number 27 is divisible by 9, it is a Harshad number.

Question 4 :

Write a program to check if a number is a buzz number. A number is said to be a buzz number if it ends with 7 or is divisible by 7.

For example,

107 is a buzz number since it ends with 7. 28 is also a buzz number since it is divisible by 7.

Question 5 :

Write a program to check if a number is an automorphic number. A number is said to be an automorphic number if the square of the number ends with the number itself.

For example,

$25^2 = 625$ = 625. Since 625 ends with 25, 25 is called an automorphic number.

Question 6 :

Write a program to check if two numbers are co-prime numbers. Two numbers are said to be co-prime numbers if they have only 1 as a common factor.

For example,

13 and 15 have only 1 as the common factor. They are both therefore co-prime numbers. Take the two numbers as input from the user and print "co-prime" if they are co-prime and "not co-prime" if they are not co-prime numbers.

Question 7 :

Write a program to check if a number is a Triangular number. A number is said to be a triangular number if it is formed by the sum of consecutive integers beginning from 1.

For Example,

1 + 2 + 3 + 4 = 10
1 + 2 + 3 + 4 + 5 + 6 + 7 = 28

Here, 10 and 28 are Triangular numbers.

Question 8 :

Write a program to check if a number is a Happy number. A number is said to be a Happy number if the number can be reduced to one if we keep adding the squares of the digits of the numbers.

For example, If we take 82,

$8^2 + 2^2 = 68$
$6^2 + 8^2 = 100$
$1^2 + 0^2 + 0^2 = 1$

Question 9 :

Write a program to find the L.C.M (Least Common Multiple) of two numbers. Take the numbers as input, find the L.C.M and display it. The L.C.M of two numbers is the smallest number that is divisible by both the numbers.

For example,

If we take 2 and 6, the smallest number that can be divided by both 2 and 6 is 6.
If we take 2 and 3, the smallest value divisible by both 2 and 3 is 6.

Question 10 :

Write a program to check if a number is a prime number. A number is said to be a prime number if it has only two factors, 1 and the number itself.

For example,

The number 11 is divisible only by 1 and 11. It is therefore a prime number.

Question 11 :

Write a program to check if a number is a Disarium number. A number is said to be a Disarium number if the sum of the power of the positions from left to right is equal to the number itself.

For example, if we take 175 → $1^1 + 7^2 + 5^2 = 1 + 49 + 125 = 175$
Therefore, 175 is a Disarium number

Question 12 :

Write a program to check if a number is a unique number. A number is said to be a unique number if all its digits are unique, that is, none of the digits of the numbers are repeated.

For example, 12654 is a unique number.
13858 is not a unique number (8 is repeated twice)

Question 13 :

A user is given two options to either check if a number is a Spy number or to find the Highest Common factor of two numbers. Write a program to accept an option and accordingly execute the program.

a. To check if a number is a spy number.
(A number is called a spy number if the sum of its digits is equal to the product of its digits. For example, if we take 123.

$$\text{Sum of digits} \rightarrow 1 + 2 + 3 = 6$$
$$\text{Product of digits} \rightarrow 1 \times 2 \times 3 = 6$$

b. To find the highest common factor of two numbers.
(The highest common factor is the largest number that can divide two numbers. For example, if we take 25 and 30. The largest number that can divide both is 5. Therefore, the highest common factor is 5)